KENBRIGHT ACTUARIAL AND FINANCIAL SERVICES LIMITED

PRICING FUNDAMENTALS

INTERNAL TECHNICAL PROCEDURES

VERSION 1.0

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KAFS POLICY

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OUR REF

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DOCUMENT PURPOSE

This document has been produced as part of the internal technical procedures for KAFS and provides guidance on pricing fundamentals.

DOCUMENT OWNERSHIP

The owner of this document is Kenbright Actuarial and Financial Services Limited.

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LIST OF ABBREVIATIONS

NAME	DEFINITION
KAFS	Kenbright Actuarial & Financial Services Limited
Valuation Date	The date at which liabilities of an insurer are determined, e.g. final day of a quarter or financial year

1. DATA REQUIREMENTS

1.1. The data required for the calculation of Frequency, Severity, Office Premium and Risk Premium are:

a) Premium Data
b) Claims Data

Premium Data

1.2. The premium data used should include all policies written by the insurer up to the valuation date.

1.3. The data items required in this file are:

a) Unique policy identifier – policy number
b) Class of business – based on the regulatory classification
c) Policy start-date
d) Policy end-date

Claims Data

Paid Claims

1.4. The claims data used includes all past and current claims settled by the insurer, whether fully or partially, over the years.

1.5. Data items required in this file are:

a) Unique claim identifier – Claim number
b) Class of business – based on the regulatory classification
c) Date of loss
d) Date of registration (Reporting date)
e) Date of payment
f) Gross claim paid

Outstanding Claims

1.6. The claims data used also includes the Outstanding claims as at the valuation date and should include the following data items:

a) Unique claim identifier – Claim number
b) Class of business – based on the regulatory classification
c) Date of loss
d) Date of registration (Reporting date)
e) Gross outstanding claim

2. FREQUENCY

What is Frequency?

2.1. Frequency answers the questions: **How often do claims occur?**

2.2. It measures the number of claims per unit exposure (e.g. insured vehicles, insured lives). Think of it as the probability of claim happening in each period.

2.3. The following formula is used to determine frequency:

$$Frequency = \frac{Count\ of\ Unique\ Claims}{Exposure\ (Unique)}$$

2.4. Below is a benchmark of the frequency results for:

a) Non-medical business

Frequency (Y2020 - Y2024)					
Business Class	**Y2020**	**Y2021**	**Y2022**	**Y2023**	**Y2024**
Aviation	2%	0%	10%	8%	5%
Engineering	12%	16%	10%	11%	11%
Fire Domestic	11%	9%	7%	8%	8%
Fire Industrial	4%	3%	3%	2%	3%
Liability	2%	2%	3%	3%	2%
Marine	5%	13%	9%	5%	4%
Miscellaneous	8%	8%	11%	10%	8%
Motor Commercial	13%	13%	11%	14%	14%
Motor Private	11%	13%	11%	12%	13%
Personal Accident	4%	5%	6%	7%	8%
Theft	2%	2%	3%	2%	1%
Workmen's Compensation	16%	40%	37%	23%	19%
Total	**9%**	**11%**	**10%**	**10%**	**10%**

b) Medical business

Claim Frequencies – Medical Business					
Benefit Type	**Y2020**	**Y2021**	**Y2022**	**Y2023**	**Y2024**
Dental	28%	28%	20%	25%	32%
Funeral	4%	2%	2%	1%	1%
Inpatient	7%	8%	8%	9%	8%
Maternity	8%	8%	4%	4%	4%
Optical	24%	25%	20%	22%	27%
Outpatient	326%	347%	323%	332%	376%

Exposure

2.5. Exposure is the unit of risk that an insurer agrees to cover, representing the potential for a claim occurring.

$$Exposure\ days = Max\{0, Min(End\ date, End\ of\ year) - Max\ (Start\ date, Start\ of\ year) + 1\}$$

$$Exposure\ in\ Years\ = \frac{Exposure\ in\ days}{(End\ of\ year - Start\ of\ year) + 1}$$

2.6. Exposure is calculated on premium data by determining how many days a policy was in force during the underwriting year.

2.7. In simple words: exposure is one unit of risk (like a car, a person or a house) that is insured and therefore has the potential to generate a claim.

2.8. Worked example:

Exposure for 2022 for a policy that commences on 01 July 2022 to 30 June 2023. We'll take:

- Calendar year = 01-Jan-2022 to 31-Dec-2022

Step 1: Identify the key dates

- **Start of year** = 01-Jan-2022
- **End of year** = 31-Dec-2022
- **Start date** = 01-Jul-2022
- **End date** = 30-Jun-2023

Step 2: Apply inner functions

- $\text{Min (End date,End of year)} = \text{Min } (30 - Jun - 2023, 31 - Dec - 2022) = 31 - Dec - 2022$
- $\text{Max (Start date,Start of year)} = Max\ (01 - Jul - 2022, 01 - Jan - 2022) = 01 - Jul - 2022$

Step 3: Calculate the difference

$$\text{31-Dec-2022} - \text{01-Jul-2022} + 1$$

Counting days between **1-Jul-2022 and 31-Dec-2022 inclusive** = **184 days**

Step 4: Apply the outer max

$$\text{Exposure days} = \max\ (0,184) = 184$$

3. SEVERITY

What is severity?

3.1. Severity answers the question: **How costly are the claims?**

3.2. It represents the average cost per claim and indicates the financial impact when a claim occurs.

3.3. The following formula is used to determine severity:

$$Severity = \frac{Total\ Gross\ Amount}{Count\ of\ Unique\ Claims}$$

3.4. Below is a benchmark of the Count of unique claims results for:

a) Non-medical business

Number of Incurred Claims (2020 - 2024)					
Business Class	**Y2020**	**Y2021**	**Y2022**	**Y2023**	**Y2024**
Aviation	1	0	5	4	3
Engineering	56	77	57	63	61
Fire Domestic	313	259	174	174	187
Fire Industrial	103	83	84	67	75
Liability	38	56	71	73	40
Marine	32	72	50	36	24
Miscellaneous	198	235	413	434	301
Motor Commercial	675	672	710	881	663
Motor Private	1,986	2,266	2,475	2,250	2,087
Personal Accident	150	221	244	205	161
Theft	35	25	43	33	15
Workmen's Compensation	240	579	577	327	271
Total	**3,827**	**4,545**	**4,903**	**4,547**	**3,888**

b) Medical business

Number of Incurred Claims– Medical Business					
Benefit Type	**Y2020**	**Y2021**	**Y2022**	**Y2023**	**Y2024**
Dental	12,031	12,967	15,634	19,020	20,531
Funeral	189	216	216	193	196
Inpatient	9,276	11,285	12,968	14,468	14,509
Maternity	2,375	2,286	2,394	2,304	1,765
Optical	10,109	11,556	15,599	17,406	17,517
Outpatient	261,126	319,294	385,712	407,362	451,046

3.5. Below is a benchmark of the severity results for:

a) Non-medical business

Average Claims Severity (2020 - 2024)					
Business Class	**Y2020**	**Y2021**	**Y2022**	**Y2023**	**Y2024**
Aviation	35,250	0	6,562,481	2,218,922	1,507,917
Engineering	328,552	594,660	567,486	780,532	1,242,478
Fire Domestic	69,161	79,248	97,596	147,755	152,044
Fire Industrial	1,491,170	2,443,961	2,610,959	1,829,667	5,227,816
Liability	789,038	1,065,758	862,926	582,547	757,829
Marine	150,666	210,755	818,849	158,595	1,699,404
Miscellaneous	607,825	132,072	116,822	194,068	158,019
Motor Commercial	757,525	608,713	545,929	599,418	466,937
Motor Private	286,545	343,031	346,987	375,770	418,441
Personal Accident	1,127,923	552,298	706,812	564,625	535,018
Theft	660,589	797,025	1,869,565	1,234,639	1,564,960
Workmen's Compensation	375,121	118,722	135,915	202,762	155,818
Total	**447,236**	**389,911**	**413,950**	**425,593**	**502,624**

b) Medical business

Average Claim Severity – Medical Business					
Benefit Type	**Y2020**	**Y2021**	**Y2022**	**Y2023**	**Y2024**
Dental	9,131	10,089	10,533	10,440	9,769
Funeral	101,529	103,515	101,713	101,954	101,462
Inpatient	153,352	160,963	160,181	146,882	166,437
Maternity	87,684	92,450	88,840	90,783	100,014
Optical	12,952	13,270	13,958	13,688	13,708
Outpatient	5,444	5,636	5,501	5,561	5,870

3.6. Why are frequency and severity important?

1. They distinguish between the "*likelihood of a claim occurring*" (Frequency) and the "*financial impact of that claim*" (Severity).
2. Multiplying them gives Risk Premium which represents the expected loss per unit of exposure. (*See Pure Risk Premium & Office Premium Chapter*)

Points to note

- Low severity and high frequency claims are common in classes of business like Motor insurance (e.g. small accidents, scratches, or windscreen repairs), Medical insurance (e.g. outpatient visits and routine check-ups), and Fire Domestic package insurance (e.g. minor household damages).
- These claims are usually small but happen very often, so the total cost can still be high for the insurer.
- High severity and low frequency claims are typical in classes of business like Fire Industrial (e.g. factory burning down), marine and aviation insurance (e.g. ship or aircraft loss).
- These claims are rare, but when they occur, the cost is very large and can have a big impact on the insurer's finances.

4. RISK PREMIUM & OFFICE PREMIUM

Risk Premium

4.1. Risk premium is also referred to as **pure premium**.

4.2. This is the expected cost of future claims, calculated based on the likelihood of claims occurring and the potential size of the loss, representing the amount an insurer requires to cover expected claims.

$$Pure\ Risk\ Premium = Frequency * Severity$$

or

$$Risk\ Premium = \frac{Total\ Gross\ Amount}{Exposure\ (Unique)}$$

Office Premium

4.3. Office Premium, also referred to as the **Gross Written Premium**, is the total amount charged to the policyholder for insurance coverage.

4.4. Office Premium incorporates not only the Risk Premium (expected cost of claims) but also additional loadings to ensure the sustainability of the insurance contract, including:

i. Expenses such as administration and commissions

ii. Profit margin for the insurer

iii. Contingency margins to allow for unexpected adverse experience

iv. Reinsurance costs, where applicable

4.5. It is calculated as risk premium/ percentage of office premium (Loss ratio)

$$Office\ Premium = \frac{Risk\ Premium}{Loss\ Ratio}$$

*Loss ratio refers to the proportion of premium used to pay claims.

Note

A lower loss ratio means that a smaller share of the premium goes toward paying claims. As a result, a larger share must be set aside for expenses, commissions, and profit. This increases the loading built into the office premium. Such that:

- If the **loss ratio is high**, it means most of the premium is being consumed by claims → less left for expenses and profit → **Office Premium must increase** to remain viable.
- If the **loss ratio is low**, fewer claims are paid relative to premium → more room for expenses/profit → **Office Premium can be lower** (relative to the risk premium).

Difference between Office Premium and Risk Premium

Aspect	Risk Premium	Office Premium (Gross Premium)
Definition	Portion of the premium that covers only the expected cost of claims.	Actual premium charged to the policyholder.
Purpose	Provides a theoretical estimate of claim costs; used as a starting point for pricing.	Ensure the insurer covers all costs and earns profit while offering sustainable products.
Inclusions	Claims cost only (e.g., death, illness, accident benefits).	Claims cost **plus** expenses (administration, acquisition, commissions), contingency margins, and profit.
Relative Size	Lower, since it excludes business overheads and margins.	Higher, because it reflects full business requirements.
Practical Use	Used by actuaries in internal calculations and pricing models.	Paid by policyholders as the final premium on their insurance contract.

5. GUIDE TO FREQUENCY & SEVERITY CALCULATION

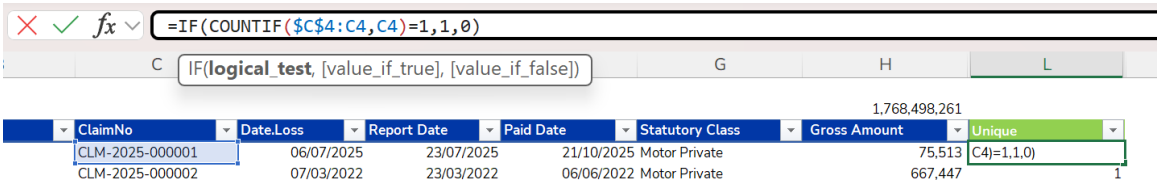
Step 1- Data

5.1. Begin by cleaning the claims and premium data to ensure accuracy, consistency, and completeness before any analysis is performed.

5.2. Confirm that entries are saved in their right formats e.g. dates, numbers, and text.

Step 2 – Calculating Severity

5.3. First, add a new column for unique claims. Unique claims are distinct claim records identified without duplication, ensuring that each claim is counted only once in the analysis.



5.4. Create tables that summarize the total gross claim amounts and the count of unique claims, categorized by loss year and statutory class.

5.5. Each statutory class should show the total gross claim amounts paid in a given loss year.

5.6. Apply the severity formula.

Step 3- Calculate frequency

5.7. First, add a new column for unique policy count. Unique policies are distinct policies records identified without duplication, ensuring that each policy is counted only once in the analysis.

5.8. Use the same uniqueness formula applied earlier. (*In step 2*)

5.9. Apply the exposure formula for exposure.

5.10. Create a table summarizing unique exposure, categorized by underwriting year and statutory class

5.11. Use the unique claims count from the severity computation.

5.12. Apply the frequency formula.

Note: Frequency is expressed as a percentage.

Step 4- Risk premium and Office premium

5.13. With all the necessary data available, create a risk premium table and calculate the values using the risk premium formula.

5.14. Once the risk premium is established, the next step is to develop an office premium table and calculate the office premium using the derived risk premium together with the applicable loss ratio.

5.15. Construct the Risk Premium table.

Using the available claims and exposure data, calculate the risk premium with the risk premium formula. The risk premium reflects the expected cost of claims per unit of exposure, without loading for expenses, commission, or profit.

5.16. Construct the Office Premium table.

Once the risk premium is determined, calculate the office premium by adjusting the risk premium with the relevant loadings (e.g., expense ratio, commission, profit margin, or target loss ratio). This step ensures the final premium charged to customers covers not only claims but also operational and business requirements.

***Assume a loss ratio of 60%**

6. APPROVAL

Thus, adopted and signed by the management and Chief Actuary of KAFS in January 2026.

APPROVED BY

Ezekiel Macharia

Chief Actuary & Managing Director – Kenbright Actuarial and Financial Services Limited

Date: January 2026

Signed:

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